

NIGHTTIME VEHICLE LIGHT DETECTION BASED ON DEEP LEARNING AND IMAGE ENHANCEMENT

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ABSTRACT

Driving vehicle at night is one of the main periods of traffic accidents. The reason for this is that the lighting at night is poor. For objects with the same direction and the opposite direction of the vehicle motion at night, only the lights can be used as the basis for discrimination, so vehicle detection becomes very difficult. In this study, under the condition that only lights are used when driving at night, the night image is converted into a Nakagami image that can enhance the light area. Then the area and features of the light are obtained by Edgebox technology. Besides, the CycleGAN is also used to convert night images into daytime images. Then we can use YOLO to detect vehicles to ensure that the obtained light area belongs to a part of the vehicle. This method can exclude interference, and identification error occurred from other light sources. Finally, by using feature fusion, SVM is applied for vehicle light classified and vehicle positioning, so that the accuracy of vehicle lights detection will not be reduced due to different weather or light conditions at night. The experimental results show that the proposed method has an 81% average precision ratio and a 78% average recall ratio for the identification of night vehicle lights.

Keywords: Light detection, image translation, nighttime vehicle detection, nighttime image enhancement.

1. INTRODUCTION

Vision is the most crucial sensory method of human beings, but statistics of driving accidents clearly show the consequences of limited vision. For example, perception problems are one of the critical reasons in every road traffic accident [1]; serious traffic accidents are prone to occur in dark and dusk situations [2]. If we consider that there is only a relatively small proportion of driving activities in dusk and darkness (about 25%), but 55% of driving accidents occur during this period, which means that night driving has a 3.67 times higher rate of accidents than day driving. According to [3], under the elimination of related factors such as fatigue and traffic density, the probability of accidents involving pedestrians in the dark has increased by 4.14 times, and the probability of accidents caused by collisions with animals has increased by 4.60 times. These statistics indicate that improving the

driving environment's perception for the vehicle and the driver is urgent action, especially in places where there are no other available light sources (for example, street lighting, etc.). Therefore, what can be done in a vehicle is to make technical improvements for its lighting system, or to improve the ability of driving at night through ADAS technology.

At present, because of the advancement of ADAS technology and the development of the self-driving car industry, how to improve the ability to sense the traffic environment when driving at night, especially the surrounding vehicles, is one of the core technology needs. For example, for self-driving cars, L3-level cars need to consider the driver as one of the passengers [4] and do not need to monitor the current road conditions at any time. Therefore, vehicles need to use various types of real-time road condition monitoring systems instead of drivers. For nighttime, when only the vehicle lights are visible, it is still necessary to know the position and speed of the vehicle in the same direction or reverse direction, and to predict the relationship early. For the advanced ADB system [5-7] while projecting sufficient illumination (i.e., using the high beam) for the driver, it is necessary to decide the vehicle that moves in different positions and carry out the corresponding light projection pattern of the vehicle headlights to avoid causing other driving glare effects.

Vehicle detection at night through computer vision technology has some different processing conditions than daytime vehicle detection, which makes nighttime vehicle detection very difficult [8]. First of all, traditional nighttime images have the characteristics of low brightness background, low definition, and high headlight brightness, which is very different from daytime images. Therefore, the effectiveness of using traditional daytime image enhancement technology on nighttime images is not obvious. In the night image, it is impossible to provide a bright vehicle appearance. The traditional characteristics of the vehicle, such as wheels, vehicle models, colors, etc., become infeasible as a method of vehicle identification and positioning. The only distinct feature of the vehicle is the vehicle lights. In the nighttime road environment, the vehicle lights moving in the opposite direction and the same direction; there are still bright objects such as street lights, advertising signs, traffic signs, reflections, etc., which

makes the main processing objects of the vehicle lights easy to be misjudged or interference.

This paper proposes a novel nighttime image enhancement technology to make the system more accurate in capturing the moving lights in opposite and same directions. In order to eliminate the interference of different light sources in the night traffic environment, deep learning technology is applied, which can obtain the imaging area of the vehicle from the night image, thus becoming beneficial auxiliary information to determine the correct position of the vehicle lights. Through the captured vehicle lights from night images and the positioning of vehicle bodies using deep learning, feature fusion can be performed on information from different sources to make the location of the vehicle lights robust, and the identified vehicle light regions have high accuracy.

2. RELATED WORKS

2.1 Light position detection

Light source detection is used to detect headlight points of the vehicle. Most methods are based on threshold technology, the most common of which is a fixed threshold [9] or adaptive threshold [7, 10]. The threshold method is processed very quickly, and it still has a good effect on the application of detecting vehicle lights, but it cannot capture the complete target object structure. The processing of light spot pairing helps to exclude the reflection caused by other light sources, and it is easy for the system to make correct decisions. However, if spots of similar size and shape are found near the candidate spots, they are likely to be recognized as the same vehicle [11-12].

The sensor-based method mainly uses sensors such as light intensity [13], LiDAR, sonar, and so on, to detect the traffic situation. However, this type of method requires a particular device, which increases the cost. Because the detected physical quantity is related to the application environment, there are limitations in its use.

2.2 Vehicle identification technology

The use of computer vision to accurately detect other vehicles on the road is still a difficult task and has been an important research topic in the past two decades [14]. The road situation is dynamic due to the changing background and lights of the vehicle. At the same time, all vehicles on the road are usually moving, so in the image obtained by the camera, the size and position of the vehicle are variable. Therefore, in the past ten years, related methods of vehicle detection and tracking have been widely explored in the literature.

In the traditional method, the primary artificial extraction features for vehicle detection are Haar [15], HOG [16] and LBP [17], and the classifiers matching using these three features mainly include support vector machines (SVM) [18], Adaboost [19] and Adaboost [20] combined with Haar feature set. Vehicle detection uses these classifier frameworks, which are mainly used for

face detection, and still finds to have good performance. The HOG function combined with the SVM classifier is the most widely used method for image recognition [21] and has achieved great success in pedestrian detection. In addition to the broad applicability of the above features and classifiers in vehicle detection tasks, a statistical model based on vertical and horizontal edge features is also proposed for vehicle detection [22] and vehicle tracking by locating taillights at night.

The characteristics of vehicles in the night environment are not prominent, so traditional car or headlight identification algorithms based on daytime images cannot effectively operate. Therefore, the vehicle detection and identification of night images mainly rely on the characteristics of the vehicle lights. At the same time, it is assumed that the high-brightness vehicle lights are used as the characteristics and recognition objects of the vehicle. Robert [23] expressed the input image as a VHS color model, and the vehicle lights had the highest ratio of brightness to saturation, and segmented and classified in pixels. Zhang et al. fed the Markov random field model through the constructed reflection intensity map, reflection suppression map, and pixel brightness, and the vehicle/light position was obtained from the pixel value [24]. For the detection and identification of multiple vehicles, it is through tracking the spots of the vehicle lights and taillights to improve the ability to distinguish the vehicles by cross-correlation [25].

The use of tracking processing is to provide light spot motion information, which is very useful when identifying vehicles at night. A standard method for detecting and tracking vehicle lights at night is to use Kalman filters [11, 18, 26]. The classification is to filter out the non-vehicle light spots because of the difficulty of identification caused by the reflection of non-light sources and street lights. Typical solutions include rule-based methods [27] and machine learning methods [28].

Many objects with high performance have been published for object detection and recognition from daytime images [29-30]. However, in night images, the contrast between the background and the objects and the overall brightness is too low, so that specific details of the vehicle, such as the edge of the vehicle, color and shape characteristics, will become unclear. Therefore, most of the known methods are to enhance the contrast, brightness, and details of the night image before feature extraction, to make accurate vehicle detection possible. For the application of ADB, in addition to capturing vehicle light in night images, it also has higher requirements for the accuracy of the measurement of its position, so the pre-processing of night images not only needs to enhance the separation of lights and background but also be able to assist in determining the location of the vehicle.

2.3 Vehicle detection based on deep learning

In recent years in the field of computer vision, due to the increasing amount of data and the rapid development of hardware, the timely performance of

deep learning vehicle detection algorithms has been compared to traditional vehicle detection algorithms based on manual features. Moreover, the detection accuracy of deep learning methods far exceeds that of traditional algorithms. Several current mainstream deep learning object detection algorithms such as R-CNN [31], R-FCN [32], AlexNet [33], RetinaNet [34] and YOLOv3 [35], etc.

A novel method is to use image conversion in deep learning technology, the unpaired image conversion method based on cycle-consistent adversarial networks [36], which can do image conversion by learning the relationship between input and output. This method can use unpaired day and night images to translate between images. At the same time, nighttime images can be translated into corresponding daytime images with a small amount of distortion, and existing algorithms that are highly efficient for daytime images can be used to detect and segment objects in the sensing environment without being affected by the low-contrast and brightness [37].

3. LIGHT AREA EXTRACTION

This paper proposes an enhancement method for night images so that the lights to be detected can be highlighted, and it is also easy to calculate the characteristics of the light area. In order to eliminate the light source similar to the vehicle lights in the night road environment, we tried to use deep learning calculations to directly convert the night images into daytime images, which can quickly obtain the imaging area of the vehicle, and at the same time become an effective discrimination mechanism for the location of the vehicle lights in the image. The next step is to merge the features obtained from these two different types of information and use SVM for the final classification of the vehicle lights. The entire processing flow is shown in Figure 1, and each processing procedure is described as follows.

3.1 Night image enhancement

The quality of the night image is usually poor, but it is an essential basis for the detection of vehicle lights. Therefore, the contrast enhancement and feature enhancement of the night image are processed first, and it is easier to obtain stable vehicle light characteristics. Referring to the method of Kuang et al. [18], first of all, the contrast enhancement of night images, especially the ability to enhance the brightness of the vehicle lights, so the color night images $I_o(R(x, y), G(x, y), B(x, y))$ is converted into a high-contrast grayscale image $I_c(x, y)$. The low-luminance night background and vehicle lights can be separated. The conversion formula is as follows:

$$I_c(x, y) = \frac{\text{Max}(R(x, y), G(x, y), B(x, y))}{255} \quad (1)$$

In order to improve the noise reduction and contrast enhancement between the vehicle lights and the background, a step function (threshold t_1) is used for

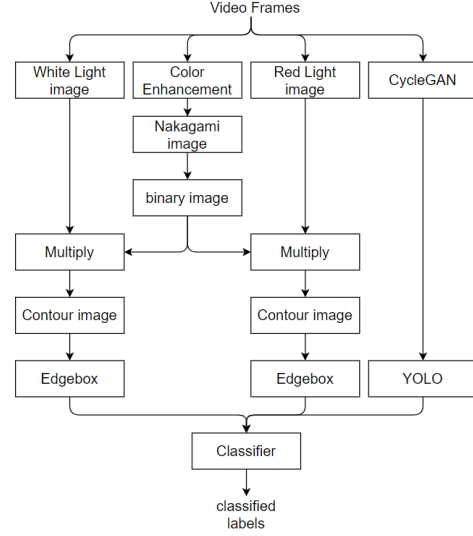


Fig. 1. Overview of the proposed light detection method.

filtering, and a grayscale image $I_F(x, y)$ with better contrast is obtained.

$$I_F(x, y) = \begin{cases} I_c(x, y), & I_c(x, y) > t_1 \\ 0, & I_c(x, y) \leq t_1 \end{cases} \quad (2)$$

Brightness is the main feature of detecting vehicle lights. In this study, the vehicle light area in the grayscale image $I_F(x, y)$ is further enhanced. The Nakagami distribution is a member of the gamma distribution group, which can be used to enhance the contrast of video content [38]. For the random variable X , its Nakagami distribution is mainly defined by two parameters:

$$\Omega = E[X^2] \quad (3)$$

$$m = \frac{\Omega^2}{E[(X^2 - \Omega)^2]} = \frac{E[X^2]^2}{V(X^2)} \quad (4)$$

Consider a pixel p in the grayscale image I_F as a random variable. Then, an $n \times n$ area W_p is built around it, and the square of the n^2 pixel values is regarded as the sampling of the pixel p , and the parameters of its Nakagami distribution can be calculated according to equations (3) and (4). Then reconstruct the Nakagami probability value corresponding to each pixel into an image form, that is, become the Nakagami image I_N . Due to the square processing of the pixel values, the image contrast is strengthened. By the readjustment from the Nakagami distribution, the brightness of vehicle lights in the Nakagami image is more apparent than other objects (road signs or trees) in the background at night, and the area is more complete.

To simplify the processing of the number of possible light areas, the Nakagami image is converted into a binary image I_B through a preset thresholding value. Figure 2 shows the results of the implementation of night image enhancement, which can be more robust by capturing the vehicle light areas of the Nakagami image.

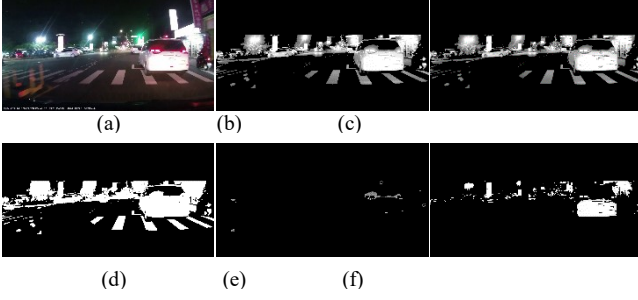


Fig. 2. The process of nighttime image enhancement: (a) the original image, (b) the enhanced image, (c) the Nakagami image, (d) the Nakagami image thresholding, (e) the red light image, (f) the white light image.

3.2 Extraction of the vehicle light area

Since the size of the vehicle light in the image is related to the distance between the vehicles and because the camera and the road surface maintain a fixed height and direction, the imaging of farther vehicle lights is higher and smaller in vertical. Assuming that the camera is installed and calibrated in advance, the FOE of the image can be kept at the center of the image, even if the optical axis of the camera is consistent with the direction of the vehicle moving. If a row value r is set to indicate the upper limit of the imaging of vehicle light, so the bright areas appearing in the area of $0 \leq y \leq r$ cannot belong to the lights. Many street lights, traffic signs, and advertisements can be filtered out as a vehicle light area.

Next, connected-components labeling is performed on the I_B , so that the processing unit of the video content is increased from pixels to regions, which is conducive to the correct discrimination of the vehicle lights. Due to the low-quality characteristics of night images, pre-processing of noise and area holes is required before the label processing of the connected components that to reduce the complexity and errors of the label processing. In this system, erosion and dilation operations of morphology are used to perform different calculations to make the processing result of the vehicle light area more complete.

For each region existing in the I_B , the edgebox [38] algorithm is used to determine the range of each bright area. The box b is represented by the smallest enclosing rectangle and can be regarded as a candidate region (ROI) of the lamp region. There are too much overlap and inclusion relationship between the bright areas, which requires further filtering and identification. Figure 3 shows the distribution of candidates for the headlight area after edgebox calculation processing after segmentation of the headlight area by night image enhancement and Nakagami images.

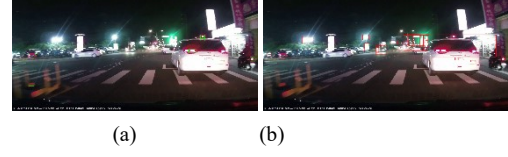


Fig. 3. The candidates of the light area by Edgebox operation. (a) the red ROIs, (b) the white ROIs.

4. VEHICLE AREA EXTRACTION

4.1 Night to day image conversion

The traditional method of directly obtaining the vehicle light area in the night image is easy to misinterpret the area. However, in the daytime image, the imaging area of the vehicle body is large, and the appearance features are apparent, so we need to find the vehicle light for simplicity and stability. Based on this idea, this study refers to the method of CycleGAN [36], which uses deep learning technology first to convert night images into day images, so that the input of night images can be converted through the model after deep learning. Furthermore, it can use the existing vehicle identification method in the image to find the location of the vehicle area corresponding to the night image.

In the real world, there is no pair of daytime and nighttime images with the same content, so the traditional GAN model has too many random mappings between the input field X and the output field Y , so there is no guarantee that the two fields are mapping correctly. The CycleGAN uses an additional Cycle consistency loss function (cycle consistency loss), thus reducing the number of random mappings and making the corresponding degree of consistency greatly improved. At the same time, the GAN model can be trained in a way that the input and output data do not need to be paired (unpaired). In this study, the collected nighttime image set is defined as X , and the collected daytime image set is defined as Y , and then we use them for training the CycleGAN model. Therefore, during the testing phase, the captured night image can be converted through the CycleGAN model to generate the corresponding daytime images. This method can also be applied to nighttime images of different weather types, and can also obtain approximate pseudo-daytime images.

In the night-to-day image processing program, the Alderley Day/Night dataset will be used to train the model on GPU 1080ti with 200 epochs to obtain a pseudo-daytime image corresponding to the nighttime input image. Figure 4 shows the result of the nighttime images captured by the on-board camera being converted into daytime images.

4.2 Vehicle detection in daytime image

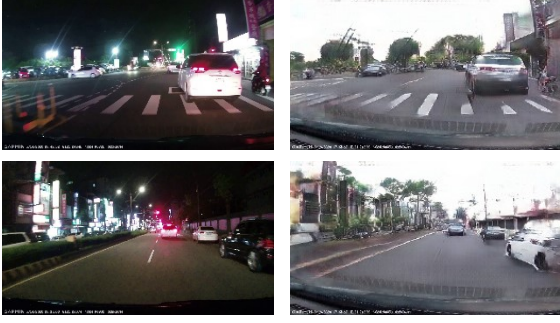


Fig. 4. Night-to-day image translation processing. The first column is the input nighttime image, and the second column is the translated pseudo daytime image output.

This study uses the YOLO algorithm for real-time vehicle image recognition for object detection based on the converted pseudo-daytime images. The characteristic of YOLO's model is that it only needs one CNN training on the image, which can determine the type and position of the object in the image, improve the recognition speed and accuracy (mAP), and show much higher performance than other deep learning models.

When the pseudo-daytime image is obtained, it will use YOLO to identify and segment the vehicle. The result is the enclosing rectangle (defined by the coordinates of the upper left corner and the height and width of the rectangle) and the confidence scores of the vehicle in the image, such as Figure 5 shows. The enclosing rectangle output will be regarded as the range in which the vehicle appears, and will be further combined with the candidate vehicle light position obtained in the previous section to achieve a more stable goal of obtaining the vehicle light position.

5. VEHICLE LIGHT CLASSIFICATION

The night image is enhanced as the Nakagami image that can obtain the position and brightness distribution information of the ROI areas related to the vehicle lights. Besides, the imaging range of the vehicle obtained through the deep learning model can be used as another type of feature of the ROI area for vehicle light classification. The corresponding possible feature sets of the ROI region of the vehicle lights are obtained, including:

- (1) U : The area ratio of the intersection area between the ROI area detected by EdgeBoxes and the vehicle area detected by YOLO. It is used to evaluate the possibility that the vehicle light belongs to the vehicle area.

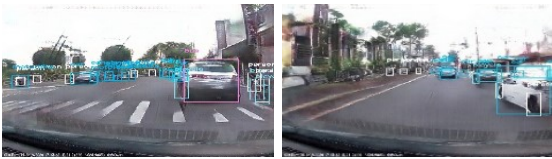


Fig. 5. Vehicle detection of pseudo-daytime images in Fig. 4 using YOLO.

- (2) H : the maximum hue value of all pixels in the central horizontal column of the ROI area. (The hue values of both white and red lights are 0), used to determine whether it is the color of vehicle lights.
- (3) D : The standard deviation of hue values on the central horizontal row of the ROI area. It is used to evaluate whether it belongs to the distribution of the typical vehicle light brightness.
- (4) Y : The row coordinate of the ROI area that is used to assess the possibility of the location of the vehicle light.
- (5) A : The area of the ROI that is used to evaluate the imaging size of regular vehicle lights

Therefore, for the i -th ROI area of the vehicle light, the feature vector for classifying vehicle light is expressed as $F_i = [U_i, H_i, D_i, Y_i, A_i]^T$.

In this study, two types of vehicle light were defined based on the vehicle moving directions (same direction and opposite direction). After feature fusion, the classification of vehicle lights is carried out by the Support Vector Machine (SVM) in supervised machine learning.

First, we establish positive and negative samples for training and label the type of vehicle lights. A training sample set of (F_i, y_i) , $i = 1, \dots, n$, is generated from the training samples, where F_i is the feature vector of the i -th sample, and y_i is its type label. Based on the principle of SVM, it is equivalent to dividing normal vectors of different types of hyperplanes in a 5-dimensional feature space. The classification separation function is transformed into an optimization problem:

$$\text{Minimize } J(W) = \frac{1}{2} \|W\|^2$$

$$\text{Subject to } y_i (W^T F_i) \geq 1, i = 1, \dots, n \quad (5)$$

6. EXPERIMENTAL RESULTS

In order to verify the correctness and accuracy of the proposed method, we collected 50 nighttime driving videos under different road and traffic conditions for training and testing purposes. All the videos are recorded through a driving recorder that is placed in front of the vehicle. Each image file contains a length of 3 minutes, so there are a total of about 150 minutes of driving records. Through the two types of vehicle lights labeling, we selected 100 RGB images of 1920×1080 , and obtained a total of 3,087 positive samples, and randomly obtained 748 negative samples, including road reflection, street lights, signboards for SVM classifier training.

The deep learning model for converting nighttime images to daytime images is based on the CycleGAN model. Randomly select unpaired 256×256 RGB images from the driving recorder video at nighttime and daytime for transfer training to obtain the deep learning model that suits for road conditions in Taiwan. For capturing the imaging range of the vehicle in the pseudo-daytime image, the existing YOLOv3 model is used to classify the vehicle. The training computer used in this study is

equipped with Intel Core i7-7700 with NVIDIA GP104 [GeForce GTX 1070].

6.1 Nakagami image and Edgebox light sources extraction

In the light detection stage, the original nighttime image is enhanced with the Nakagami distribution according to the method in Section 3. The Nakagami image can enhance the difference in brightness between the background and the lights. Figure 6 shows the image enhancement effect under different road conditions, which enhances the possible light source area to be brighter. At the same time, the background part tends to be dark, so the segmentation accuracy of the light area can be improved. Through the obtained boundary of the light source area, we can quickly obtain the possible area of the vehicle lights by Edgebox technology, and it is regarded as the ROI area of the vehicle light.

6.2 Night-to-day image conversion

In the Night-to-day image conversion processing, referring to the method of cycle-consistent adversarial networks [36] (CycleGAN), the Day/Night Alderley data set [40] is used to split the night and day data into training data sets: train_night, test_night, train_day, test_day. Then use the nighttime images of test_night as input into CycleGAN, and use the 200 epochs recommended in the [36] to train with 1080ti. The completed CycleGAN model is then imported using the image I_{in} of test_night, and the image I_{out} is generated that the result is shown in Fig. 7. The experimental results show that the position of the vehicle can be reliably converted. Although the color of the car is incorrect, it does not detract from the required vehicle body position. For different environments (urban or highway) and lighting conditions (day, night, dusk, and dawn) are shown in the example in Figure 7. We can find that most of the vehicle features can be translated at different road traffic conditions.

6.3 Vehicle detection

The purpose of the vehicle position and range detection is to define the position and range where the vehicle lights can exist. After the night-to-day image



Fig. 6. The detection of the vehicle light areas by Edgebox operation in nighttime image. The odd column is the original images and the even column is the ROIs. The white vehicle light is indicated by the red box and the red vehicle light is the green box.



Fig. 7. Night-to-day image translation processing under different road conditions.



Fig. 8. Vehicle area detection of the pseudo daytime images in Fig.7 using YOLO.

translation, the daytime image is obtained, and YOLO is used in the image for vehicle detection. This study uses darkflow installation, darkflow supports YOLOv3, and the input images are pseudo daytime images. As shown in Figure 8, YOLO can use in the pseudo daytime image and makes enclosing rectangle marked with the car, bus, bicycle, which is the expected vehicle category.

6.4 Performance evaluation

Using the feature fusion calculation on the vehicle light ROI obtained from the night image and the vehicle area obtained from the pseudo daytime image, we can get five features for each vehicle light ROI, and then classify the vehicle light category by the SVM. The discriminating results of the two types of vehicle lights in different directions are shown in Table 1. For the vehicle lights of the same direction, including white and red lights, the precision is 80%, and the recall is 90%; for the vehicle lights of the reverse direction, the precision is 83%, and the recall is 68%. Since the reverse light has only white light, it is easy to be mistaken as in the same direction, resulting in a lower recall. In the future, a mechanism for tracking vehicle lights will be introduced to strengthen the ability to recognize the direction of movement.

Table 1. Performance evaluation of vehicle light classification

	Type	Classification		Recall
		same	reverse	
Ground truth	same	2,724	312	0.897
	reverse	701	1,477	0.678
Precision		0.795	0.825	

7. CONCLUSION AND FUTURE WORK

A deep learning technique combining night image enhancement technology and pseudo daytime image generation is proposed to make the vehicle light easily obtained from the nighttime image. The vehicle range obtained from the pseudo daytime image is adapted to filter the light source in the nighttime image. This deep

learning technology uses unpaired input and output image learning to complete the conversion of nighttime images for making the feature extraction of vehicles and lights more robust. Using the fusion between two different types of data, SVM is used to classify the vehicle lights in different moving directions. Experimental results show that the recognition rate of night lights can be more than 80%. At the same time, it can have stable classification results for traffic conditions with different levels of complexity.

This research is part of the ADB system development. In the future, the output pattern of the headlight matrix will be calculated through the obtained image position of the vehicle lights to achieve the function of masking light in the ADB system. The results of this research have a high market relevance to the automotive lamp industry. The success of system development is expected to generate attractive automotive lamp display application requirements.

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